

FINAL YEAR ENGINEERING PROJECT

ORBITAL TIG WELDING FOR INDUSTRIAL PIPING

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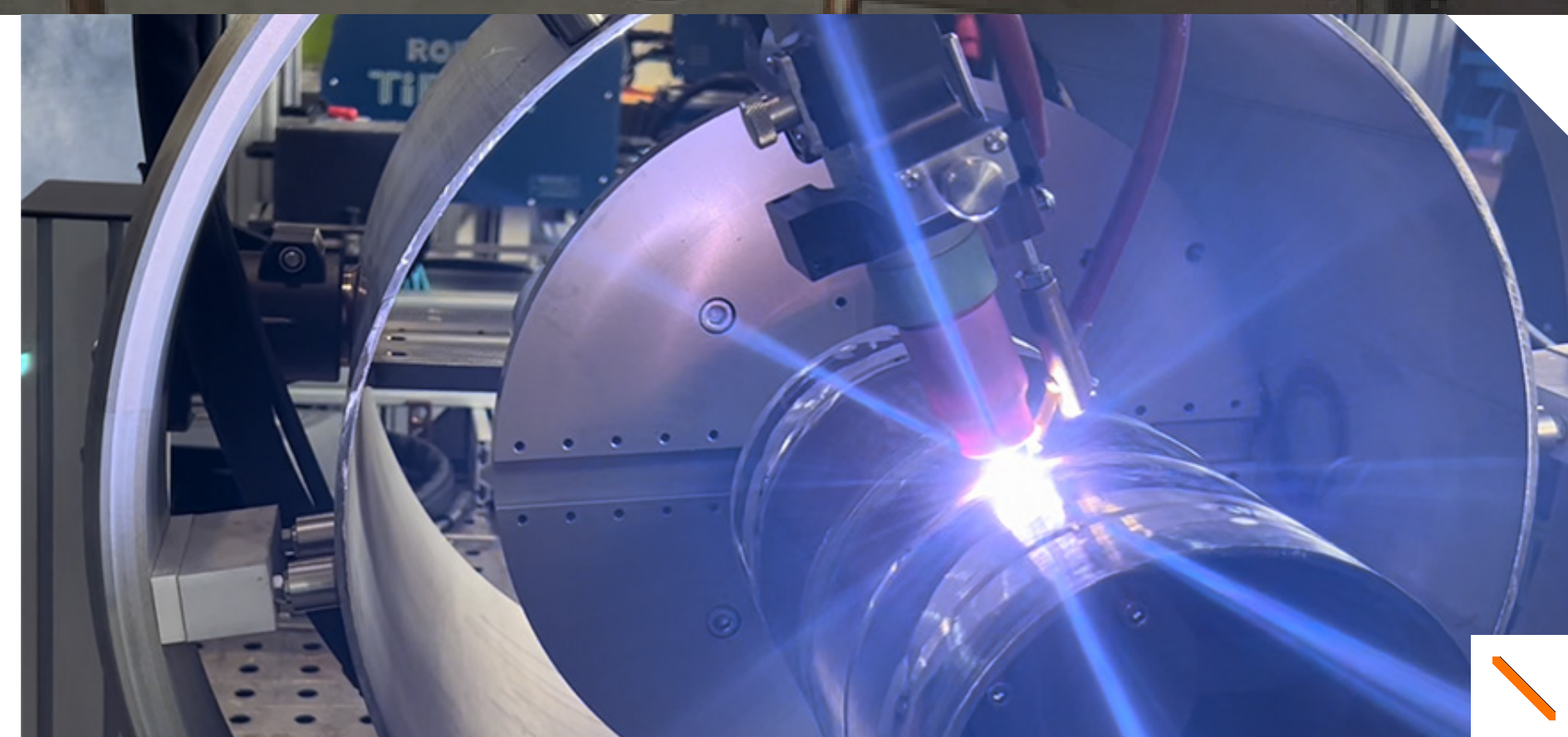


THE COMPANY

-  **CIMAT SARTEC**, FOSELEV Group. Industrial piping and maintenance.
-  Chemical, petrochemical, manufacturing and steel industries.
-  Boilermaking, steel structures, piping and pressure equipment.
-  1155 boulevard du Grand Terme, 30133 Les Angles.

MY ROLE

Apprentice project engineer: estimating, supplier consultation, planning, fabrication follow-up and inspection, through to installation on site.



ORBITAL TIG WELDING

-  Torch carried right around a pipe held in **fixed position**.
-  **Pulsed current**: the molten pool stays under control.
-  **Multiple angular sectors**, one set of parameters for each welding position.
-  Enclosed head, argon shielding and **back purging** of the root.

SCOPE STUDIED

Austenitic stainless steel 304L and 316L, small and medium diameters, the everyday production of the workshop.



BEFORE AND AFTER THE STUDY

-  **Before**: orbital welding was an idea nobody had costed. No technical assessment, no plan, no offer.
-  **After**: a complete decision file. Technical assessment, compliance path with ISO 14732, supplier offer, costing.
-  **On the table**: a quantified recommendation, ready for management to decide.
-  **Expected in the workshop**: a bead that repeats, roughly half the welding time, parameters logged by the machine.

STATUS

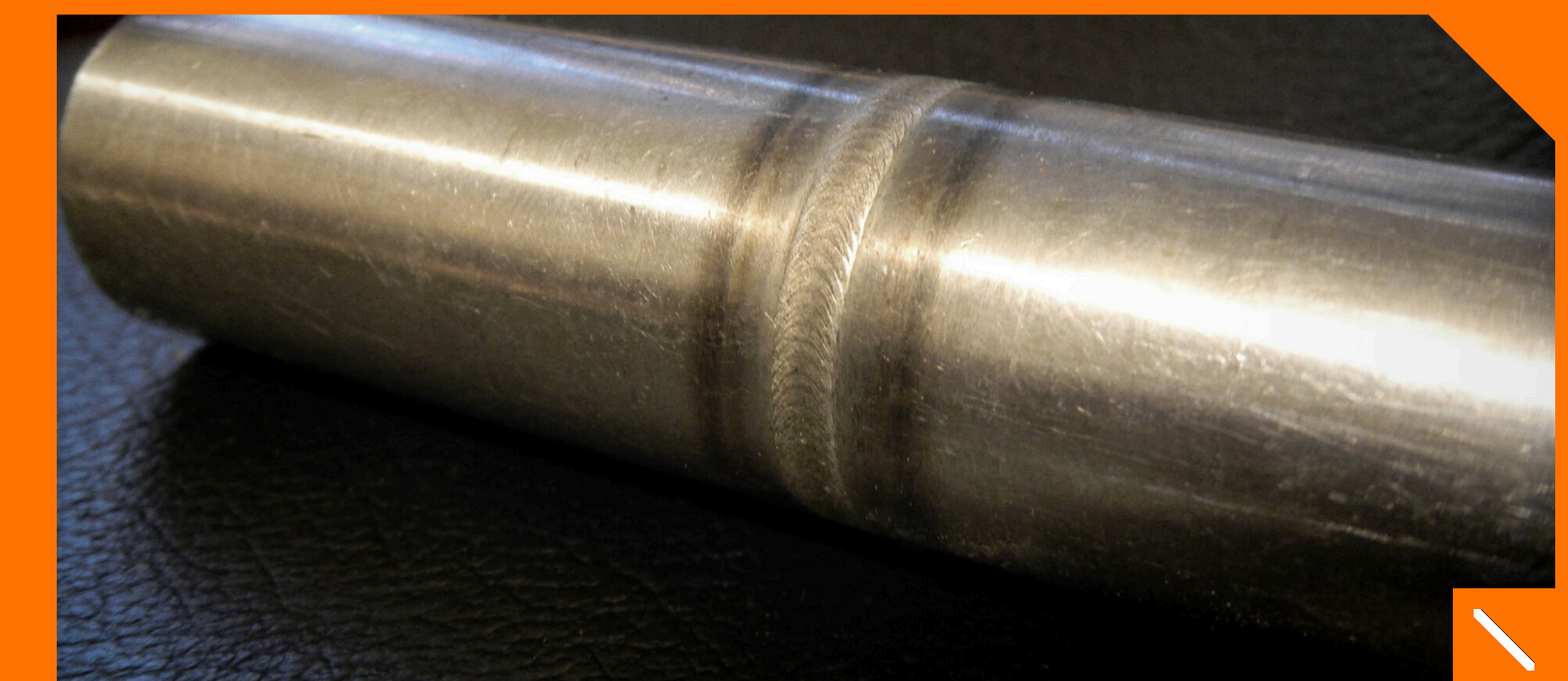
The process is **not installed**. The study was delivered, and the decision sits with the management team.

WORTH THE INVESTMENT

That is the conclusion of the study, not a decision already taken. Welding times still have to be confirmed by measurement in the workshop.

IF IT GOES AHEAD

-  Welding on site, alongside the teams.
-  Open head fed with wire for large diameters and thick walls.
-  A library of welding programmes that gains value over time.



Finished orbital weld on 316L: 4X4 Blazer 1776, Wikimedia Commons, CC BY 2.0. Other photos: FOSELEV and equipment manufacturers.